CISS350: Data Structures and Advanced Algorithms
Quiz q10901

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Here's how you describe (in text) for a binary tree with a heap tree structure. For this tree



you write

```
{10, 6, 7, 5, 2, 0, 3, 1}
```

i.e., list the values using breadth-first traversal (like what we did in class).

Q1. Describe (using text) the following maxheap after you have inserted 9 to it.



Answer:

```
{10, 6, 7, 5, 2, 0, 3, 1, 9}
{10, 6, 7, 9, 2, 0, 3, 1, 5}
{10, 9, 7, 6, 2, 0, 3, 1, 5}
```

Q2. Describe (using text) the following *min*heap after you have inserted 0 to it.



Answer:

```
[1, 3, 2, 6, 4, 8, 5, 9, 7, 0]
[1, 3, 2, 6, 0, 8, 5, 9, 7, 4]
[1, 0, 2, 6, 3, 8, 5, 9, 7, 4]
[0, 1, 2, 6, 3, 8, 5, 9, 7, 4]
```

Q3. Describe (using text) the following maxheap after you have performed extract-root on it.

```
              (10)
           /        \
        (6)          (7)
       /   \        /   \
    (5)     (2)  (0)     (3)
   /
(1)
```

Answer:

```
[10, 6, 7, 5, 2, 0, 3, 1]

[1, 6, 7, 5, 2, 0, 3]
[7, 6, 1, 5, 2, 0, 3]
[7, 6, 3, 5, 2, 0, 1]
```

Q4. Complete the following maxheap insert function where `std::vector< int >` is used for the array representation of the maxheap.

Answer:

```
#include <iostream>
#include <string>
#include <vector>

std::ostream & operator<<(std::ostream & cout, const std::vector< int > & v)
{
    cout << '[';
    std::string delim = "";
    for (auto & e: v)
    {
        cout << delim << e; delim = ", ";
    }
    cout << ']';
    return cout;
}

int PARENT(int i)
{
```

```
  return (i - 1) / 2;
}

int LEFT(int i)
{
  return 2 * i + 1;
}

int RIGHT(int i)
{
  return 2 * i + 1;
}

void maxheap_insert(std::vector< int > & v, int x)
{
  v.push_back(x);

  int n = v.size() - 1;
  while(n > 0 && v[n] > v[PARENT(n)]
  {
    int temp = v[n];
    v[n] = v[PARENT(n)];
    v[PARENT(n)];
    n = PARENT(i);
  }
}

int main()
{
    std::vector< int > heap;
    std::vector< int > data {5, 3, 1, 2, 4, 6, 0};
    for (auto & e: data)
    {
        maxheap_insert(heap, e);
        std::cout << heap << '\n';
    }
    return 0;
}
```

Instructions

In the file `thispreamble.tex` look for

```
\renewcommand\AUTHOR{}
```

and enter your email address:

```
\renewcommand\AUTHOR{jdoe5@cougars.ccis.edu}
```

(This is not really necessary since alex will change that for you when you execute `make`.) In your bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

Enter your answers in `main.tex`. In the bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

For each question, you'll see boxes for you to fill. For small boxes, if you see

```
1 + 1 = \answerbox{}.
```

you do this:

```
1 + 1 = \answerbox{2}.
```

`answerbox` will also appear in "true/false" and "multiple-choice" questions.

For longer answers that need typewriter font, if you see

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
\end{answercode}
```

you do this:

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
int x;
\end{answercode}
```

`answercode` will appear in questions asking for code, algorithm, and program output. In this case, indentation and spacing is significant. For program output, I do look at spaces and newlines.

For long answers (not in typewriter font) if you see

```
What is the color of the sky?
\begin{answerlong}
\end{answerlong}
```

you can write

```
What is the color of the sky?
\begin{answerlong}
The color of the sky is blue.
\end{answerlong}
```

A question that begins with "T or F or M" requires you to identify whether it is true or false, or meaningless. "Meaningless" means something's wrong with the question and it is not well-defined. Something like "$1 + 2 = 4$" is either true or false (of course it's false). Something like "$1+_2 = 4$?" does not make sense.

When writing results of computations, make sure it's simplified. For instance write 2 instead of $1 + 1$.

Higher level classes.

For students beyond 245: You can put LaTeX commands in `answerlong`.

More examples of meaningless statements: Questions such as "Is $42 = 1+_2$ true or false?" or "Is $42 = \{2\}^{\{3\}}$ true or false?" does not make sense. "Is $P(42) = \{42\}$ true or false?" is meaningless because $P(X)$ is only defined if X is a set. For "Is $1 + 2 + 3$ true or false?", "$1 + 2 + 3$" is well–defined but as a "numerical expression", not as a "proposition", i.e., it cannot be true or false. Therefore "Is $1 + 2 + 3$ true or false?" is also not a well-defined question.

More examples of simplification: When you write down sets, if the answer is $\{1\}$, do not write $\{1, 1\}$. And when the values can be ordered, write the elements of the set in ascending order. When writing polynomials, begin with the highest degree term.

When writing a counterexample, always write the simplest.